

Cart Abandonment Pattern Analysis

Exploratory Data Analysis (EDA) Summary

Dataset: February 2020 e-commerce behavior data. The analysis used the complete February file, not a sample. The raw file contained 55,318,565 events and was processed in chunks because the file was about 7.15 GB.

1. What was the purpose of the EDA?

The goal was to understand why customers who added products to their cart did not complete a purchase. We first built a clean cart-session view, then checked whether abandonment changed with product category, brand, price, time of day, and customer session behavior.

2. Initial dataset inspection

We first loaded a small 1,000-row sample only to understand the structure of the file. This was not used as the analysis sample. It showed 9 columns: event_time, event_type, product_id, category_id, category_code, brand, price, user_id, and user_session.

The initial sample contained 926 views, 65 carts, and 9 purchases. This confirmed that the dataset was an event-level clickstream dataset rather than a ready-made cart table.

3. Full February dataset check

We then scanned the full February file in chunks. Chunking was necessary because the raw file was too large to safely load into memory in one go.

Event / metric	Result
Total events	55,318,565
Views	51,232,669
Cart events	2,885,608
Purchase events	1,200,288
Unique users	4,233,206
Unique sessions	13,544,595

Why this mattered: before doing any EDA, we needed to know the real size of the data and confirm that views, carts, and purchases were present in the expected event types.

4. Building the cart-level analytical dataset

The raw file contained tens of millions of events, but the business question was specifically about carts. So we retained cart-event information such as event time, product, category, brand, price, and user session. Purchase information was also used at session level to identify whether a cart session eventually converted.

This reduced the data to a much smaller and more useful dataset for cart-abandonment analysis.

5. Duplicate check

We checked the cart-event data for exact duplicate rows. The purpose was to make sure the same event was not being counted multiple times and artificially increasing cart activity.

Measure	Rows
Raw cart-event rows	2,885,608

Exact duplicates	229,080
Remaining unique cart rows	2,656,528

Result: 2,656,528 unique cart-event rows were used for the detailed cart-level analysis.

6. Missing category values

Category_code was missing for 203,714 cart-event rows, or about 7.67%. We did not delete these rows because a missing category is still a real cart event. Instead, missing categories were labelled as 'Unknown'. The same idea was used for missing brands when doing brand analysis.

7. Session-level funnel

The next step was to move from individual events to sessions. A session was treated as converted when it contained both a cart event and a purchase event. A cart session without a purchase was treated as abandoned.

Metric	Sessions
View sessions	13,217,465
Cart sessions	1,660,110
Purchase sessions	754,013
View + cart sessions	1,383,989
View + cart + purchase sessions	698,497
Cart + purchase sessions	700,091
Abandoned cart sessions	960,019

Overall abandonment rate: 57.83% Cart-to-purchase conversion: 42.17%.

This became the baseline for every later comparison. For example, a category with 80% abandonment was clearly above the overall 57.83% level.

8. Category analysis

We compared abandonment across product categories. We first looked at high-volume categories so that a category with only a handful of sessions would not dominate the result. We also checked categories with at least 1,000 sessions for a more reliable comparison.

Category	Cart sessions	Abandonment
construction.tools.light	657,113	47.64%
Unknown	139,227	68.91%
sport.bicycle	118,267	61.29%
electronics.clocks	63,778	61.06%
apparel.shoes	56,099	63.49%
electronics.smartphone	4,263	93.81%
computers.notebook	N/A	82.85%
air_conditioner	N/A	81.49%
painting	N/A	80.22%
chair	N/A	77.97%

Finding: abandonment was not the same across categories. Smartphones, notebooks, air conditioners, painting, and chairs showed much higher abandonment than the overall 57.83% rate. Construction tools had a lower rate despite having by far the largest session volume.

9. Brand analysis

We repeated the same idea for brands. We used a minimum of 1,000 cart sessions so that very small brands would not produce misleading extreme percentages.

Brand	Sessions	Converted	Abandoned	Abandonment
silverlit	1,831	41	1,790	97.76%
nokian	2,521	296	2,225	88.26%
rivertoys	1,672	204	1,468	87.80%
yokohama	2,780	457	2,323	83.56%
viatti	1,665	284	1,381	82.94%
micelin	1,167	205	962	82.43%
toyo	1,001	185	816	81.52%
maxxis	1,139	218	921	80.86%
triangle	4,071	816	3,255	79.96%
powertrac	1,146	250	896	78.18%
skad	2,046	459	1,587	77.57%
cordiant	6,792	1,548	5,244	77.21%
tunga	1,446	337	1,109	76.69%
electrolux	1,018	240	778	76.42%
lego	1,079	259	820	76.00%

Finding: some brands had extremely high abandonment. However, many of the highest-abandonment brands were tire/automotive brands, so the brand result may partly reflect the category or product type. Brand alone should not be treated as the cause.

10. Price analysis

We first checked the price distribution to understand what price levels were common before creating bands.

Statistic	Value
Rows	2,656,528
Mean	\$291.55
Median	\$172.44
Q1	\$71.15
Q3	\$360.09
Minimum	\$0
Maximum	\$2,574.07

We then created price bands: \$0–50, \$50–100, \$100–200, \$200–350, \$350–500, \$500–1,000, and \$1,000+.

Price band	Cart sessions	Abandonment
\$0–50	385,121	65.64%
\$50–100	222,059	60.03%
\$100–200	477,608	50.69%
\$200–350	328,903	50.29%
\$350–500	136,406	53.42%

\$500–1,000	248,869	51.83%
\$1,000+	94,093	53.09%

Finding: lower-priced cart items had higher abandonment. The \$0–50 group had 65.64% abandonment, compared with roughly 50–53% for most higher-price groups. This suggests price level is associated with abandonment, but price alone does not explain the behavior.

11. Hour-of-day analysis

We extracted the hour from each session's first cart event and calculated abandonment by UTC hour. The purpose was to see whether customers behaved differently at different times of day.

Hour	Cart sessions	Abandonment
00	8,057	62.68%
01	14,387	66.37%
02	29,803	62.62%
03	57,061	57.79%
04	82,589	56.24%
05	98,948	56.18%
06	109,235	56.10%
07	113,811	56.02%
08	116,802	55.87%
09	115,198	55.95%
10	112,768	56.22%
11	105,525	57.06%
12	98,847	56.93%
13	92,761	57.94%
14	91,593	59.35%
15	90,923	61.15%
16	83,749	61.38%
17	75,151	61.71%
18	62,827	60.38%
19	42,973	58.87%
20	25,669	57.82%
21	15,332	56.67%
22	9,236	54.16%
23	6,865	56.04%

Finding: abandonment was lower during roughly 05:00–12:00 UTC and higher around 14:00–18:00 UTC. The difference was around 5–6 percentage points. This is a time-of-day association, not proof that the hour itself causes abandonment.

12. Cart size / number of products in a session

We also counted how many different cart products were present in a cart session. This was meant to test whether larger carts behaved differently from smaller carts.

Products in cart	Cart sessions
1	1,484,011
2	139,468
3	25,332
4–5	9,305
6–10	1,874

11+	119
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Important: the session counts were calculated, but the abandonment-rate calculation for these product-count bands was not completed. Therefore, no abandonment conclusion should be drawn from this section. The 11+ group was also very small at only 119 sessions.

13. Repeat-session behavior

The final completed EDA looked at how many sessions a user had across the dataset and whether cart abandonment changed with that history. Users were grouped into 1 session, 2 sessions, 3 sessions, 4–5 sessions, and 6+ sessions.

User session count	Cart sessions	Converted	Abandoned	Abandonment
1	470,701	162,747	307,954	65.42%
2	305,300	119,573	185,727	60.83%
3	207,307	82,921	124,386	60.00%
4–5	245,038	101,436	143,602	58.60%
6+	431,764	233,414	198,350	45.94%

Finding: users with only one recorded session had 65.42% abandonment, while users with 6+ sessions had 45.94%. This is a large difference. It suggests that repeat users behave differently from one-time users and may be more likely to complete a purchase.

14. What factors affected cart abandonment the most?

The EDA did not use a machine-learning feature-importance model. So 'most affected' here means the factors that showed the clearest and largest differences in observed abandonment rates during our grouped analysis.

Factor	Evidence from EDA	What it suggests
User/session history	1 session: 65.42% vs 6+: 45.94%	Repeat users were much more likely to convert.
Product category	Several categories were 77–94% abandoned	Product type had a strong association with abandonment.
Brand	Top examples reached 76–98%	Some brands/products had very high abandonment; category mix may
Price	\$0–50: 65.64% vs \$100–350: about 50%	Lower-priced carts showed higher abandonment.
Time of day	About 55.9% around 05–12 UTC vs 59–62% around 14–18 UTC	Abandonment changed meaningfully by time of day.

15. Final EDA conclusion

The strongest practical pattern in the EDA was user history: repeat users, especially users with 6+ sessions, had much lower cart abandonment than users with only one session. Product characteristics were also important. Some categories and brands had abandonment rates far above the overall 57.83% level. Price showed a clear difference too, with the lowest price band having the highest abandonment. Time of day also mattered, with higher abandonment during the afternoon/evening UTC period.

The safest business interpretation is that cart abandonment is not driven by one single factor. It changes with customer familiarity, the type of product being considered, product/brand mix, price level, and time of day. These are observed relationships from EDA, so they should not be described as proof that any one factor directly causes abandonment.

16. What this EDA gives us for the next stages

The EDA identified the patterns. The next stage is SQL, where the same business questions can be answered using clean aggregations, filters, CASE statements, and CTEs/window functions only where they genuinely add value. Power BI can then turn the final metrics and findings into an interactive business dashboard.

Note: This report intentionally explains the purpose and result of each meaningful analysis rather than describing every individual Python syntax line.